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Estimating the True Population at Risk: Under-Five Malaria Mortality, Denominator Uncertainty, and Child Survival Investment Priorities in Zamfara State, Nigeria

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Received: 10.02.2026
Revised: 17.05.2026
Accepted: 19.05.2026
Published: 24.05.2026

Abstract

Malaria remains a leading cause of mortality among children under five years of age in Nigeria, yet uncertainty in target population estimates and the coexistence of multiple child health vulnerabilities limit the effectiveness of intervention planning. This study developed a geospatial framework to quantify underfive malaria mortality rates, evaluate denominator uncertainty, and identify child survival investment priorities across Zamfara State, Nigeria. Monthly LGA-level underfive malaria deaths from 2014–2024, household microcensus data, the WorldPop 2025 underfive population raster, malaria burden indicators, immunization coverage, and healthcare access metrics were integrated within a geographic information system. Annual underfive malaria deaths increased from 221 in 2014 to 706 in 2024, with a mean annual burden of 334 deaths. The number of predicted deaths from underfive malaria cases for 2025 ranged from fewer than 10 cases to 129 deaths, with Gusau, Talata Mafara, Gummi, Maru, and Kaura Namoda exhibiting the highest mortality burdens. The household-enumerated and raster-derived underfive populations numbered 838,969 and 848,590, respectively, differing by 9,621 children (1.15%) at the state level; however, the LGA-level discrepancies ranged from –55.4% to +54.5%, with a mean absolute difference of 25.1%. The Child Survival Gap Index identified Gusau, Talata Mafara, Maru, Bungudu, and Kaura Namoda as priority LGAs where high malaria mortality, treatment gaps, immunization deprivation, weak healthcare access, and denominator uncertainty converge. These findings provide an evidence-based framework for geographically targeted malaria and child health interventions in resource-constrained settings.

Keywords: Underfive malaria mortality; Population denominator uncertainty; Household enumeration; Child survival gap index; Geospatial health prioritization.

1. Introduction

Malaria remains one of the leading causes of morbidity and mortality among children under five years of age globally, with sub-Saharan Africa accounting for the overwhelming majority of cases and deaths [1]. Nigeria alone contributes to approximately one quarter of the global malaria burden and continues to experience substantial child mortality attributable to malaria despite decades of intervention efforts [2]. In 2023, Nigeria accounted for 26.8% of global malaria cases and 30.9% of malaria deaths, underscoring the need for improved strategies for targeting and evaluating malaria control programmes [1]. Although interventions such as seasonal malaria chemoprevention (SMC), insecticide-treated nets (ITNs), rapid diagnostic testing, and artemisinin-based combination therapies have reduced malaria transmission in many settings, progress remains uneven across regions and population groups [3].

Accurate estimation of the target population is fundamental to malaria program planning, implementation, and impact evaluation. Estimates of intervention coverage, commodity requirements, cost-effectiveness, and lives saved depend critically on the denominator used to define the population at risk [4]. In many low- and middle-income countries, programme planners rely on gridded population products, census projections, or administrative estimates because recent household enumeration data are often unavailable [5]. Among these datasets, WorldPop and related gridded population datasets have become widely used owing to their high spatial resolution and broad geographic coverage [6]. However, these products are generated through statistical modeling and may not adequately represent local demographic heterogeneity, especially in areas characterized by rapid population growth, dispersed settlements, seasonal migration, or conflict-related displacement [7].

Recent studies have reported substantial discrepancies between modeled and operationally enumerated populations used for vaccination, malaria control, and humanitarian interventions [8,9]. These discrepancies can have important consequences for programme performance. Overestimation of the target population may lead to inefficient allocation of resources and underestimation of intervention coverage, whereas underestimation may result in inadequate commodity procurement and failure to reach vulnerable populations [10]. Consequently, there is growing recognition that uncertainty in population denominators represents a major challenge for malaria programme design and evaluation [11].

Moreover, child survival is influenced by multiple interacting factors in addition to malaria mortality alone. Children living in areas with poor access to healthcare, low immunization coverage, and weak health systems are more likely to experience adverse health outcomes [12]. Spatial inequalities in healthcare access and immunization coverage have been documented across Nigeria, with northern states exhibiting persistently lower vaccination rates and higher childhood mortality rates than southern regions do [13]. Therefore, approaches that simultaneously consider malaria burden, treatment gaps, immunization deprivation, and healthcare accessibility may provide a more comprehensive understanding of child survival vulnerability and help prioritize interventions more effectively [14].

Geospatial approaches offer powerful tools for integrating diverse datasets and identifying spatial patterns of health vulnerability [15]. Previous studies have used geographic information systems (GISs), remote sensing, and spatial modeling to map malaria risk, estimate treatment accessibility, and evaluate healthcare inequities [16,17]. However, relatively few studies have explicitly examined how uncertainty in underfive population denominators influences estimates of malaria burden and child survival vulnerability. Furthermore, integrated frameworks that combine malaria mortality, denominator uncertainty, immunization deprivation, and healthcare accessibility remain scarce, particularly at subnational scales in Nigeria.

Zamfara State, located in northwestern Nigeria, represents an important setting in which to address these knowledge gaps. The state experiences high malaria transmission, persistent childhood mortality, and substantial variation in healthcare access and population distribution across local government areas (LGAs). Reliable evidence on underfive mortality patterns, denominator uncertainty, and child survival vulnerability is essential for optimizing malaria interventions and informing resource allocation in the state.

Therefore, the present study developed a geospatial framework to quantify underfive malaria mortality rates, evaluate uncertainty in underfive population denominators, and identify child survival investment priorities across Zamfara State. Specifically, the study aimed to (i) model and predict underfive malaria mortality rates for 2025 via monthly mortality records from 2014–2024; (ii) quantify discrepancies between household-enumerated and raster-derived underfive population estimates and evaluate denominator uncertainty; and (iii) develop a child survival gap index that integrates the malaria mortality burden, treatment gaps, immunization deprivation, healthcare access, and denominator uncertainty. The results reveal substantial spatial heterogeneity in malaria mortality and population estimation, identify priority LGAs where multiple dimensions of vulnerability converge, and provide an evidence-based framework for geographically targeted malaria and child health interventions.

2. Materials and methods

2.1 Study area

Zamfara State is located in northwestern Nigeria between latitudes approximately 10°15′N and 13°30′N and longitudes 5°00′E and 7°15′E (Figure 1). The state covers an area of approximately 38,418 km² and is administratively divided into 14 local government areas (LGAs): Anka, Bakura, Birnin Magaji, Bukkuyum, Bungudu, Gummi, Gusau, Kaura Namoda, Maradun, Maru, Shinkafi, Talata Mafara, Tsafe, and Zurmi. Gusau serves as the state capital and the principal urban center.

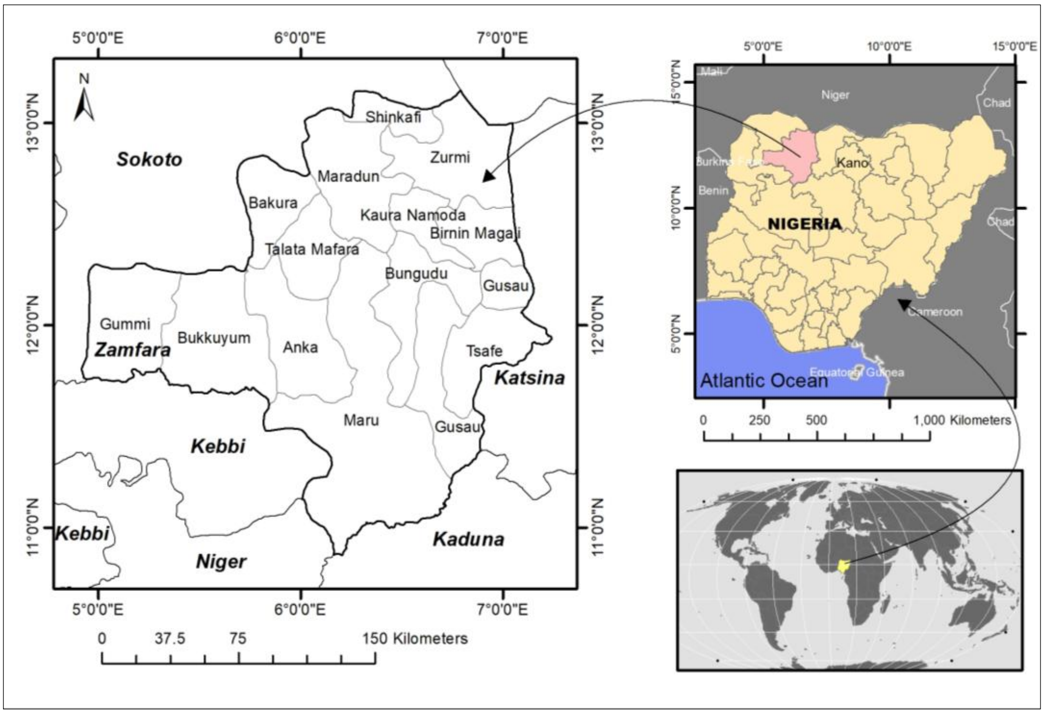

Figure 1. Location of Zamfara State in Nigeria and the distribution of the 14 local government areas (LGAs) included in this study. The map was produced by the authors via R version 4.5.1, and administrative boundary data were obtained from OCHA Nigeria.

The state lies within the Sudan and northern Guinea savanna ecological zones and is characterized by a tropical continental climate with distinct wet and dry seasons. The rainy season typically extends from May to October, whereas the dry season lasts from November to April. The mean annual rainfall ranges from approximately **700 to 1,100 mm**, with higher rainfall occurring in the southern parts of the state. The mean annual temperatures generally range between **25°C and 32°C**, with maximum temperatures often exceeding 40°C during the hot dry season.

Malaria transmission in Zamfara State is highly seasonal and is caused predominantly by *Plasmodium falciparum*. The transmission intensity increases substantially during and immediately after the rainy season because of favorable conditions for mosquito breeding and survival. Despite ongoing malaria control interventions, including insecticide-treated nets, seasonal malaria chemoprevention, and case management with artemisinin-based therapies, the state continues to experience a substantial malaria burden among children under five years of age. Furthermore, variations in healthcare accessibility, immunization coverage, settlement patterns, and socioeconomic conditions contribute to marked spatial heterogeneity in child health outcomes across LGAs.

The state was selected for this study because it represents a high-burden malaria setting where operational household microcensus data, malaria surveillance records, gridded population estimates, and healthcare accessibility datasets are available simultaneously. This combination of datasets provides a unique opportunity to investigate underfive malaria mortality rates, evaluate uncertainty in target population denominators, and identify geographically targeted priorities for improving child survival outcomes.

2.2 Study design

This study adopted an LGA-level geospatial modeling framework to estimate underfive malaria mortality risk, quantify denominator uncertainty, and identify child survival investment gaps across Zamfara State, Nigeria. The analysis was structured around three objectives: first, to model and predict 2025 underfive malaria mortality rates via monthly death surveillance records from 2014–2024; second, to compare household-enumerated, raster-derived, and administrative underfive population denominators; and third, to develop a child survival gap index that integrates mortality risk, malaria treatment gaps, immunization deprivation, healthcare access, and denominator uncertainty.

2.2 Data used

Table 1 summarizes the datasets used in this study, including malaria surveillance records, household microcensus data, gridded population estimates, malaria transmission metrics, healthcare accessibility indicators, and socioeconomic proxies. Monthly underfive malaria deaths from RHIS/DHIS2 were used to model historical mortality trends and predict the 2025 mortality burden. Household microcensus and WorldPop datasets provided independent estimates of the underfive population for denominator discrepancy analysis. Malaria transmission intensity was characterized via Malaria Atlas project indicators, while health facility locations and nighttime light data were incorporated to represent healthcare access and socioeconomic development, respectively. Administrative boundary datasets were used for spatial aggregation, zonal statistics, and cartographic visualization. Together, these datasets enabled the integrated assessment of underfive malaria mortality, denominator uncertainty, and child survival vulnerability across Zamfara State, Nigeria.

Table 1 Data Summary

Data Name	Data Type	Purpose	Source
Monthly Under-Five Malaria Deaths (2014–2024)	Vector (LGA, monthly time series)	Historical mortality trends and prediction of 2025 underfive malaria mortality	RHIS/DHIS2 [18]
Household-Enumerated Under-Five Population	Aggregated LGA Indicator	Operational denominator for mortality rates and denominator comparison	Settlement Microcensus [19]

Raster-Derived Under-Five Population (2025)	Raster (100 m)	Modeled denominator for comparison with household enumeration	WorldPop [21]
Settlement Count	Aggregated LGA Indicator	Characterization of settlement distribution	Settlement Microcensus [19]
Household Count	Aggregated LGA Indicator	Population contextualization	Settlement Microcensus [19]
RI Zero-Dose Children	Aggregated LGA Indicator	Assessment of immunization deprivation	Settlement Microcensus [19]
RI Zero-Dose Rate (U5)	Derived Indicator	Measurement of childhood immunization vulnerability	Settlement Microcensus [19]
Not Vaccinated Children	Aggregated LGA Indicator	Assessment of immunization exclusion	Settlement Microcensus [19]
Not Vaccinated Rate (U5)	Derived Indicator	Quantification of immunization deprivation	Settlement Microcensus [19]
Fully Vaccinated Rate (U5)	Derived Indicator	Assessment of immunization performance	Settlement Microcensus [19]
Severe Malaria Cases	Vector (LGA)	Severe malaria burden	RHIS/DHIS2 [18]
Severe Malaria Treated with Artesunate	Vector (LGA)	Assessment of treatment coverage	RHIS/DHIS2 [19]
Treatment Gap	Derived LGA Indicator	Quantification of untreated severe malaria burden	Derived
Accessibility-Adjusted Treatment Gap Index (AATGI)	Composite Indicator	Assessment of severe malaria treatment deficiencies	Derived
Health Facilities	Vector	Healthcare service availability	GRID3 [20]
Health Facilities per 100,000 Population	Derived Indicator	Healthcare accessibility	Derived
PfPR 2024	Raster	Malaria parasite prevalence	Malaria Atlas Project [22]
PfIR 2024	Raster	Malaria infection intensity	Malaria Atlas Project [22]
PfMR 2024	Raster	Malaria mortality burden	Malaria Atlas Project [22]
PfIC 2024	Raster	Clinical malaria incidence	Malaria Atlas Project [22]
PfMC 2024	Raster	Clinical malaria burden	Malaria Atlas Project [22]
VIIRS Nighttime Lights 2025	Raster	Proxy for socioeconomic development and urbanization	NASA/NOAA VIIRS [23]
Administrative Boundaries (LGA and State)	Vector	Spatial aggregation, zonal statistics, and mapping	OCHA Nigeria [24]

Abbreviations: HBTGI = Hidden Burden Treatment Gap Index; AASMTGI = Artesunate-Adjusted Severe Malaria Treatment Gap Index; RI = Routine Immunization; U5 = Under-Five Children; PfPR = *Plasmodium falciparum* Parasite Rate; PfIR = *Plasmodium falciparum* Infection Rate; PfMR = *Plasmodium falciparum* Mortality Rate; PfIC = *Plasmodium falciparum* Incidence Cases; PfMC = *Plasmodium falciparum* Malaria Cases; RHIS = Routine Health Information System; DHIS2 = District Health Information System 2; GRID3 = Geo-Referenced Infrastructure and Demographic Data for Development; VIIRS = Visible Infrared Imaging Radiometer Suite.

2.3 Underfive malaria mortality model

Monthly LGA-level malaria deaths among children under five years of age were converted from wide to long formats and aggregated annually for each LGA. For LGA $i$and year t, annual deaths were calculated as follows:

$$D_{i,t} = \sum_{m=1}^{12} d_{i,m,t} \quad (1)$$

where $d_{i,m,t}$ represents the number of reported underfive malaria deaths in month m. Annual trends from 2014–2024 were assessed via Mann–Kendall trend statistics and Sen's slope. The 2025 mortality burden was predicted for each LGA via autoregressive integrated moving average forecasting, with a three-year moving mean used where model fitting was unstable. The predicted underfive malaria mortality rate was calculated as follows:

$$MR_{i,2025} = \frac{\widehat{D}_{i,2025}}{U5_i} \times 1000 \quad (2)$$

where $\widehat{D}_{i,2025}$ is the predicted number of underfive malaria deaths in 2025 and where $U5_i$ is the household-enumerated underfive population.

2.4 Denominator discrepancy analysis

To evaluate the uncertainty in target population estimation, two underfive population denominators were compared: a household-enumerated underfive population derived from settlement microcensus data and a raster-derived underfive population obtained from the 2025 WorldPop gridded underfive population dataset at 100 m spatial resolution. The household-enumerated counts were aggregated from settlement points to the LGA level, whereas the raster-derived underfive population was extracted for each LGA via zonal summation.

The absolute denominator discrepancy for LGA i was calculated as:

$$\Delta_i = U5_i^{raster} - U5_i^{enum} \quad (3)$$

where $U5_i^{raster}$ is the raster-derived underfive population and where $U5_i^{enum}$ is the household-enumerated underfive population.

The relative denominator discrepancy was computed as:

$$\%\Delta_i = \frac{U5_i^{raster} - U5_i^{enum}}{U5_i^{enum}} \times 100 \quad (4)$$

Positive values indicate that raster-derived estimates exceed household-enumerated counts, whereas negative values indicate lower raster-derived estimates. To quantify the uncertainty in the operational denominators, an absolute denominator uncertainty score was calculated by normalizing the absolute percentage discrepancy to a 0–100 scale:

$$DUS_i = N(|\ \%\Delta_i\ |) \quad (5)$$

where $N(\cdot)$ denotes min–max normalization. Larger values of DUS_i indicate greater disagreement between operationally enumerated and modeled underfive populations, suggesting increased uncertainty in denominator estimates used for malaria intervention planning. LGAs were subsequently classified into five uncertainty categories: very low, low, moderate, high, and very high.

2.5 Child Survival Gap Index

A composite child survival gap index was developed to identify LGAs where child mortality risk, malaria treatment gaps, immunization deprivation, weak healthcare access, and denominator uncertainty overlap. All indicators were normalized to a 0–100 scale:

$$N(X_i) = 100\left(\frac{X_i - X_{min}}{X_{max} - X_{min}}\right) \quad (6)$$

For protective indicators, inverse normalization was applied:

$$N^{-1}(X_i) = 100 - N(X_i) \quad (7)$$

The mortality component was calculated as follows:

$$MC_i = 0.70\, N(MR_{i,2025}) + 0.30\, N(\widehat{D}_{i,2025}) \quad (8)$$

The malaria treatment-gap component was calculated as follows:

$$TGC_i = 0.60\, N(TG_i) + 0.40\, N(AATGI_i) \quad (9)$$

where TG_i represents the severe malaria treatment gap and where $AATGI_i$ represents the accessibility-adjusted treatment gap index.

The immunization deprivation component was calculated as follows:

$$IDC_i = 0.55\, N(RIZD_i) + 0.30\, N(NV_i) + 0.15\, N^{-1}(FV_i) \quad (10)$$

where $RIZD_i$ is the RI zero-dose rate among children under five years of age, NV_i is the vaccination rate, and FV_i is the fully vaccinated rate.

The access weakness component was calculated as follows:

$$AWC_i = 0.60\, N^{-1}(HF_i) + 0.40\, N^{-1}(NTL_i) \quad (11)$$

where HF_i is the health facility density per 100,000 people and where NTL_i is the nighttime light intensity.

The final child survival gap index was calculated as follows:

$$CSGI_i = 0.30\, MC_i + 0.25\, TGC_i + 0.20\, IDC_i + 0.15\, AWC_i + 0.10\, DUS_i \quad (12)$$

The final score was rescaled to 0–100 and classified into five priority groups: very low, low, moderate, high, and very high. Sensitivity analysis was conducted via an equal-weight index:

$$CSGI_i^{EW} = \frac{MC_i + TGC_i + IDC_i + AWC_i + DUS_i}{5} \quad (13)$$

Rank differences between the weighted and equal-weight indices were used to evaluate the robustness of LGA prioritization.

2.6 Spatial analysis and mapping

All datasets were harmonized to WGS84 and analyzed at the LGA level. Five population raster values were extracted via zonal summation. Mortality predictions, denominator discrepancies, and CSGI outputs were joined to LGA polygons for spatial visualization. Maps were produced for the predicted underfive malaria deaths, mortality rates, denominator uncertainty, CSGI scores, priority classes, operational gap profiles, and sensitivity rankings. All analyses were conducted in R via sf, terra, dplyr, forecast, trend, Kendall, ggplot2, and patchwork.

3. Results

3.1 Predicted Under-Five Malaria Mortality and Spatial Heterogeneity in Zamfara State

The mortality of underfive malaria cases in Zamfara State exhibited marked temporal and spatial heterogeneity from 2014–2024. At the state level, annual deaths increased from 221 in 2014 to a peak of 706 in 2024, with a mean annual mortality burden of 334 deaths, indicating a substantial increase in malaria-associated child mortality over the study period. The monthly mortality records revealed pronounced interannual variability, with recurrent peaks during high-transmission periods and an increasing burden in recent years (Figure 2a, b).

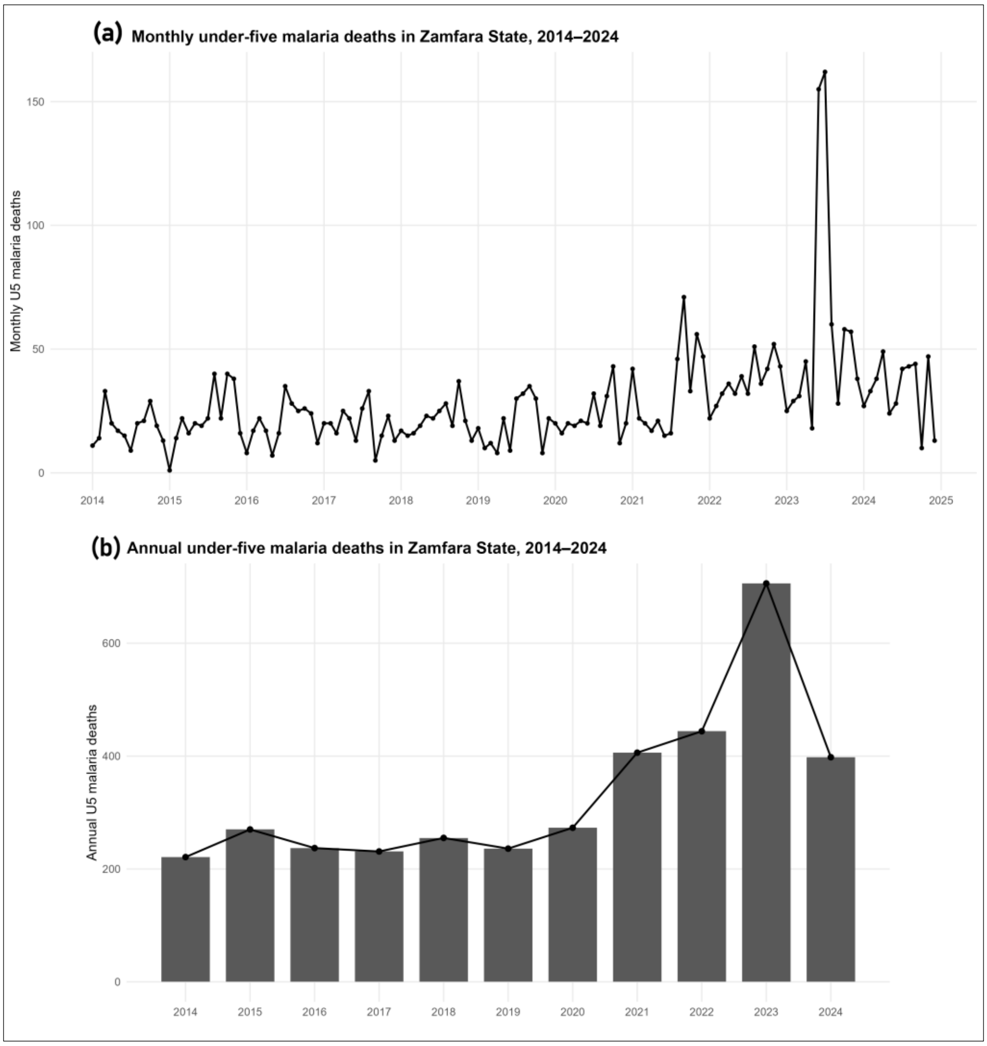

Figure 2. Temporal dynamics of underfive malaria mortality in Zamfara State from 2014–2024: (a) monthly underfive malaria deaths and (b) annual underfive malaria deaths.

At the LGA level, the cumulative number of underfive malaria deaths from 2014–2024 varied substantially across the 14 LGAs. Total deaths ranged from fewer than 50 cases in low-burden LGAs to more than 700 deaths in the highest-burden areas. The Mann–Kendall trend analysis revealed heterogeneous mortality trajectories, including significant increasing, significant decreasing, and nonsignificant trends. Positive Sen slopes were observed in several LGAs, suggesting sustained increases in malaria mortality over time, whereas other LGAs exhibited stable or declining patterns (Figure 3d).

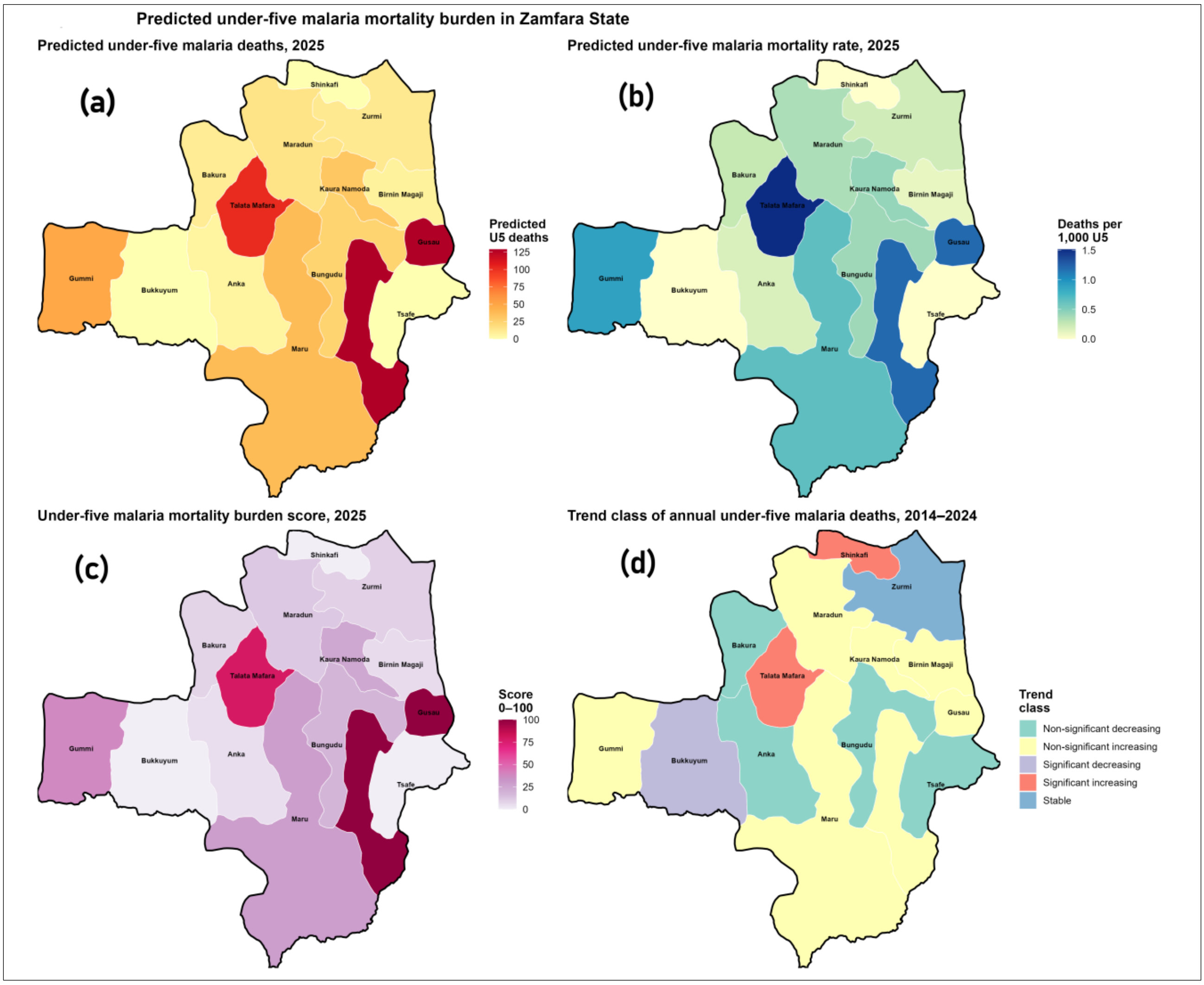

Figure 3. Spatial patterns of underfive malaria mortality rates in Zamfara State for 2025: (a) predicted underfive malaria deaths; (b) predicted underfive malaria mortality rates per 1,000 underfive population; (c) mortality burden score (0–100); and (d) underfive malaria mortality trend classes derived from Mann–Kendall and Sen's slope analyses.

Spatially, the predicted mortality rates displayed clear geographic clustering (Figure 3b). LGAs with high predicted mortality rates corresponded closely with areas exhibiting high mortality burden scores (Figure 3c), indicating persistent hotspots of underfive malaria mortality. The mortality burden score, normalized to a 0–100 scale, highlighted a gradient from very low-risk LGAs to high-priority areas requiring intensified malaria control and child health interventions. The mortality trend classes further demonstrated that high-burden LGAs are not uniformly improving, with several areas continuing to experience increasing mortality trends despite ongoing malaria control efforts (Figure 3d).

Forecasting for 2025 revealed considerable spatial disparities in predicted underfive malaria mortality rates. The number of predicted deaths ranged from fewer than 10 cases in low-burden LGAs to 129 deaths in the LGA group. Gusau was projected to experience the highest mortality burden (129 deaths), followed by Talata Mafara (102 deaths), Gummi (48.7 deaths), Maru (37.5 deaths), and Kaura Namoda (31.6 deaths) (Figure 4). These LGAs also presented an elevated severe malaria burden, substantial treatment gaps, and relatively low health facility density, suggesting that mortality risk is shaped by both disease transmission intensity and health system constraints.

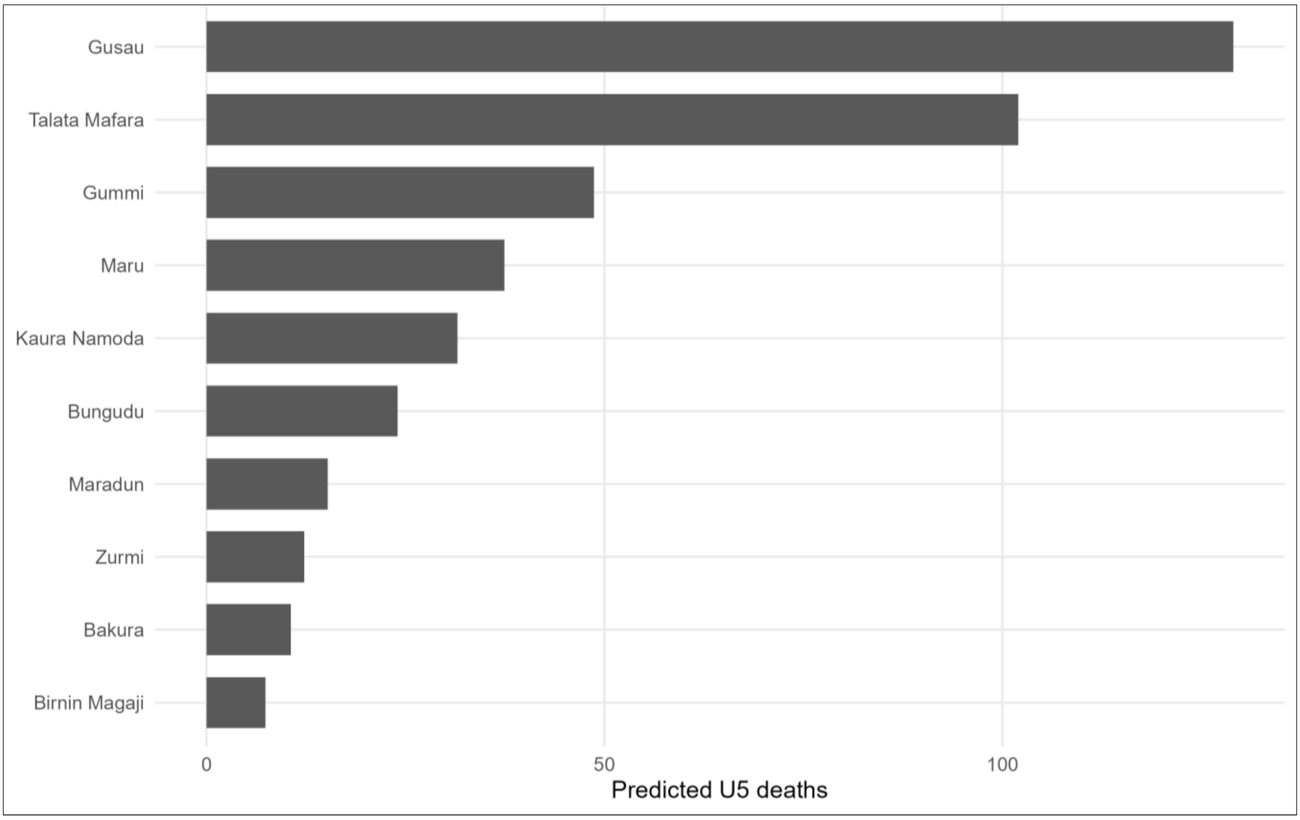

Figure 4. Top ten LGAs ranked by predicted underfive malaria deaths in 2025. The bars represent predicted deaths, highlighting LGAs with the greatest projected malaria mortality burden among children under five years of age.

Overall, the results demonstrate that underfive malaria mortality in Zamfara State is spatially concentrated and temporally dynamic, with a small number of LGAs accounting for a disproportionately large share of the predicted 2025 mortality burden. These findings provide an evidence base for geographically targeted investments aimed at reducing malaria-related child deaths and improving child survival outcomes.

3.2 Denominator Discrepancies and Uncertainty in Underffive Population Estimation

Substantial differences were observed between household-enumerated and raster-derived underfive population estimates across Zamfara State. The household enumeration identified a total of 838,969 underfive children across the 14 LGAs, whereas the raster-derived estimate yielded 848,590 children, representing an overall excess of 9,621 children or 1.15% relative to the enumerated population. Although the aggregate discrepancy at the state level was modest, considerable spatial heterogeneity was evident at the LGA level.

The spatial distributions of household-enumerated and raster-derived underfive populations are presented in Figure 5a and Figure 6b, respectively. Both datasets identified Gusau, Talata Mafara, Bungudu, and Maru as major population centers with large underfive populations. However, the magnitude and direction of the discrepancies varied considerably across the LGAs (Figure 5c). Raster-derived estimates exceeded household-enumerated counts in several LGAs, whereas in others, the raster substantially underestimated the enumerated population.

The percentage difference between raster-derived and household-enumerated underfive populations ranged from –55.4% to +54.5% (Figure 5c). The largest underestimation was observed in Birnin Magaji, where the raster estimate was 55.4% lower than the household-enumerated count, followed by Shinkafi (–51.4%). In contrast, the raster substantially overestimated underfive populations in Bungudu (+54.5%), Maru (+53.4%), and Gusau (+53.1%). Across all LGAs, the mean absolute percentage discrepancy was 25.1%, whereas the median absolute discrepancy was 14.3%, indicating that discrepancies between operationally enumerated and modeled underfive populations are nontrivial and geographically uneven.

The denominator uncertainty score, derived from the normalized absolute percentage discrepancy, further highlighted spatial inequalities in population estimation (Figure 5d). The highest uncertainty scores were observed in Birnin Magaji (100.0), Bungudu (96.7), Maru (94.4), Gusau (93.9), and Shinkafi (92.4), suggesting substantial disagreement between household-enumerated and raster-derived denominators in these LGAs. Conversely, LGAs such as Kaura Namoda and Anka presented relatively low uncertainty scores, indicating closer agreement between the two estimation approaches.

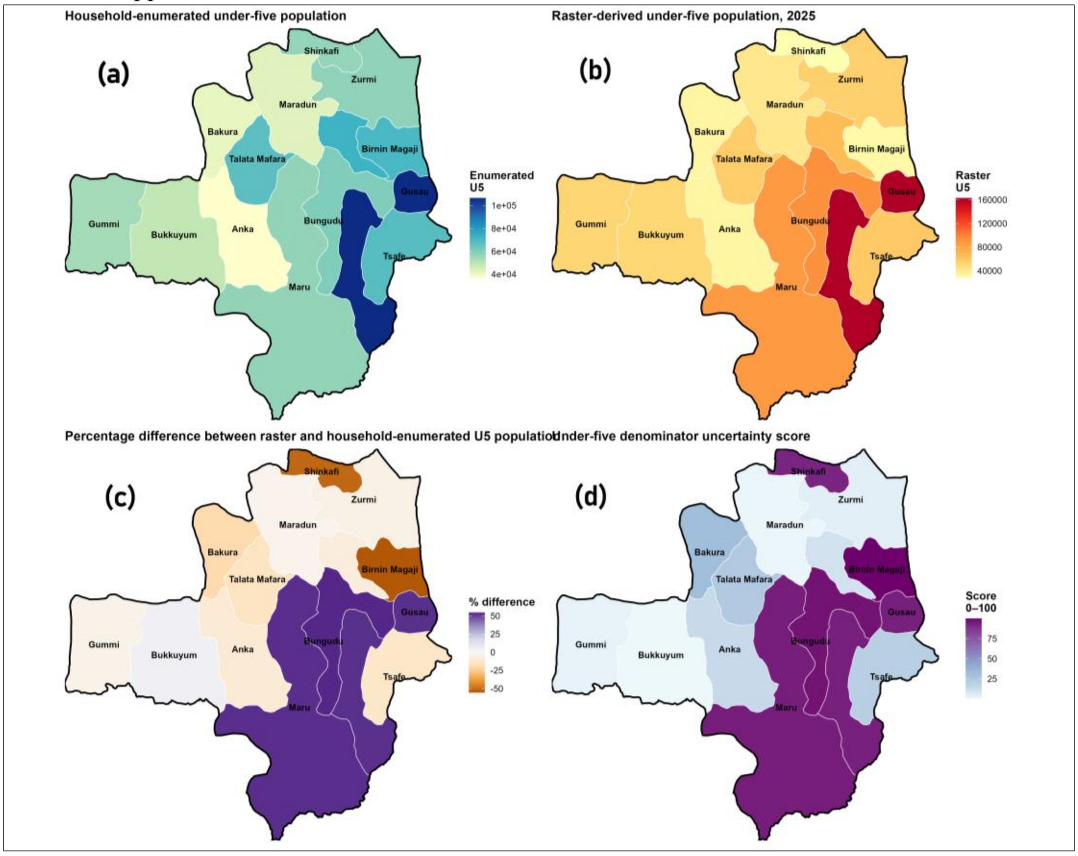

Figure 5. Comparison of household-enumerated and raster-derived underfive populations across Zamfara State: (a) household-enumerated underfive population aggregated from settlement microcensus data; (b) raster-derived underfive population extracted from the 2025 WorldPop underfive raster; (c) percentage difference between raster-derived and household-enumerated underfive populations; and (d) denominator uncertainty score normalized to a 0–100 scale.

The raster agreement classes revealed three distinct patterns of denominator concordance (Figure 6). Several LGAs exhibited approximate agreement between household-enumerated and raster-derived underfive populations, whereas others showed pronounced overestimation or underestimation by the raster model. The coexistence of these contrasting patterns across neighboring LGAs suggests that a single population modeling approach may not adequately capture local demographic heterogeneity and settlement structures.

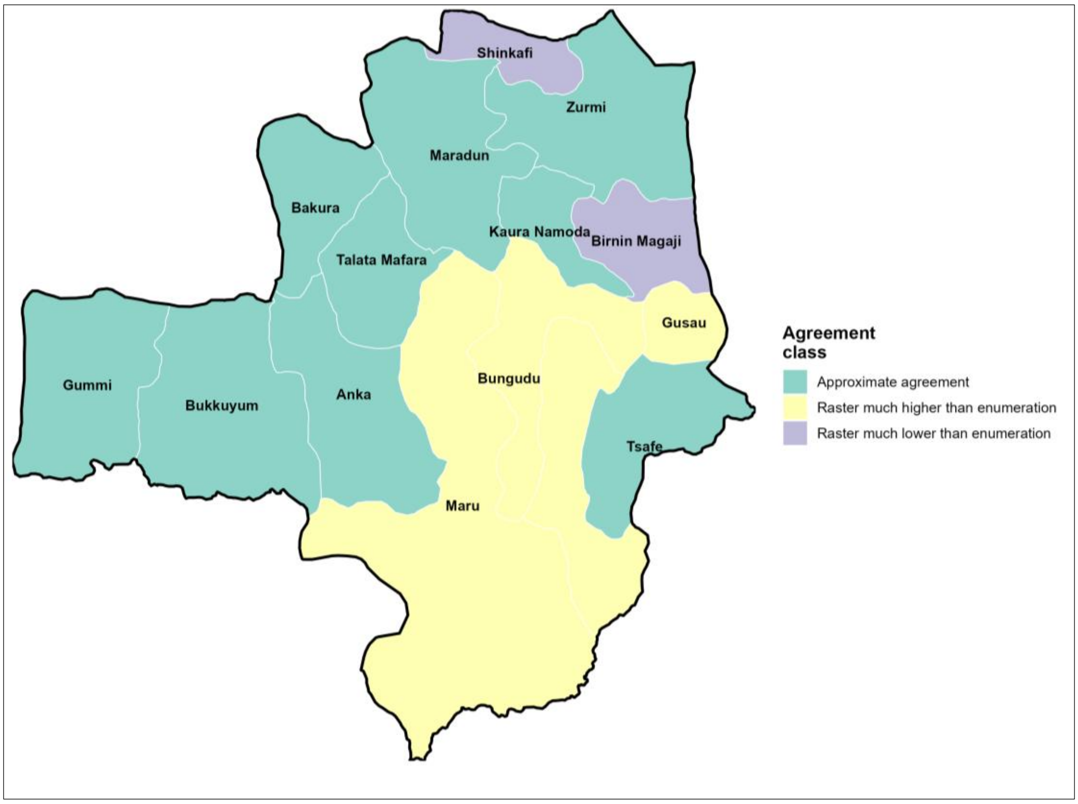

Figure 6. Raster agreement classes showing the spatial concordance between household-enumerated and raster-derived underfive population estimates across Zamfara State.

The relationships between household-enumerated and raster-derived underfive populations are illustrated in Figure 8. Although a generally positive association was observed, substantial deviations from the one-to-one agreement line indicate systematic local discrepancies.

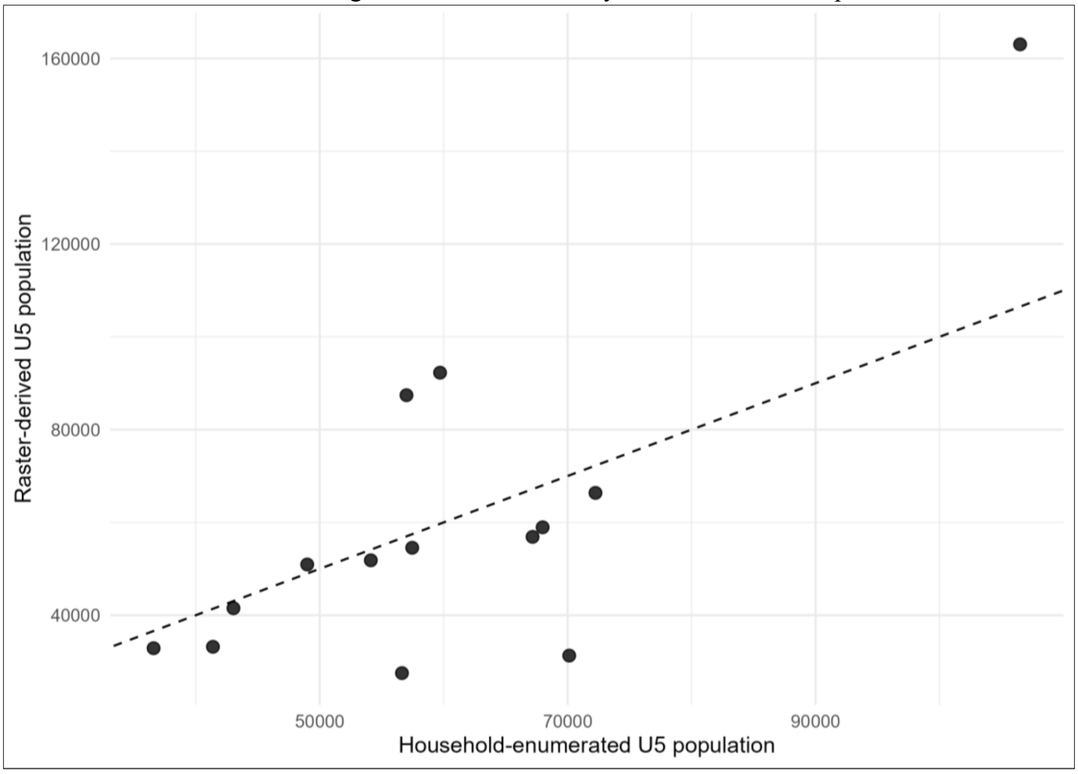

Figure 7. Scatter plot showing the relationship between household-enumerated and raster-derived underfive population estimates at the LGA level.

Overall, the results demonstrate that raster-derived population estimates reproduce the broad spatial distribution of underfive populations across Zamfara State but exhibit substantial local discrepancies. These findings underscore the importance of incorporating operational household enumeration into malaria program planning, as uncertainty in target population estimates may directly influence intervention coverage, resource requirements, and assessments of program effectiveness.

3.3 Child Survival Gap Index and Investment Prioritization

The Child Survival Gap Index (CSGI) revealed pronounced spatial inequalities in child survival vulnerability across Zamfara State, reflecting the combined effects of malaria mortality burden, treatment gaps, immunization deprivation, healthcare access, and denominator uncertainty. The composite CSGI scores ranged from 0 to 100, indicating a wide gradient of vulnerability among the 14 LGAs. The mean contributions of the mortality, treatment gap, immunization deprivation, access weakness, and denominator uncertainty components were 27.6, 28.6, 38.8, 41.0, and 47.1, respectively, suggesting that uncertainty in population denominators and weak healthcare access are important determinants of child survival gaps across the state.

The spatial distribution of the CSGI identified Gusau as the most vulnerable LGA, with a maximum score of 100 (Figure 8a). Other high-priority LGAs included Talata Mafara, Maru, Bungudu, and Kaura Namoda, which consistently ranked among the top five LGAs. These LGAs exhibited combinations of elevated predicted underfive malaria mortality, substantial treatment gaps, high RI zero-dose prevalence, and weak health-system indicators. Conversely, LGAs such as Anka, Bakura, and Zurmi presented comparatively lower CSGI scores, indicating relatively lower levels of overlapping vulnerability.

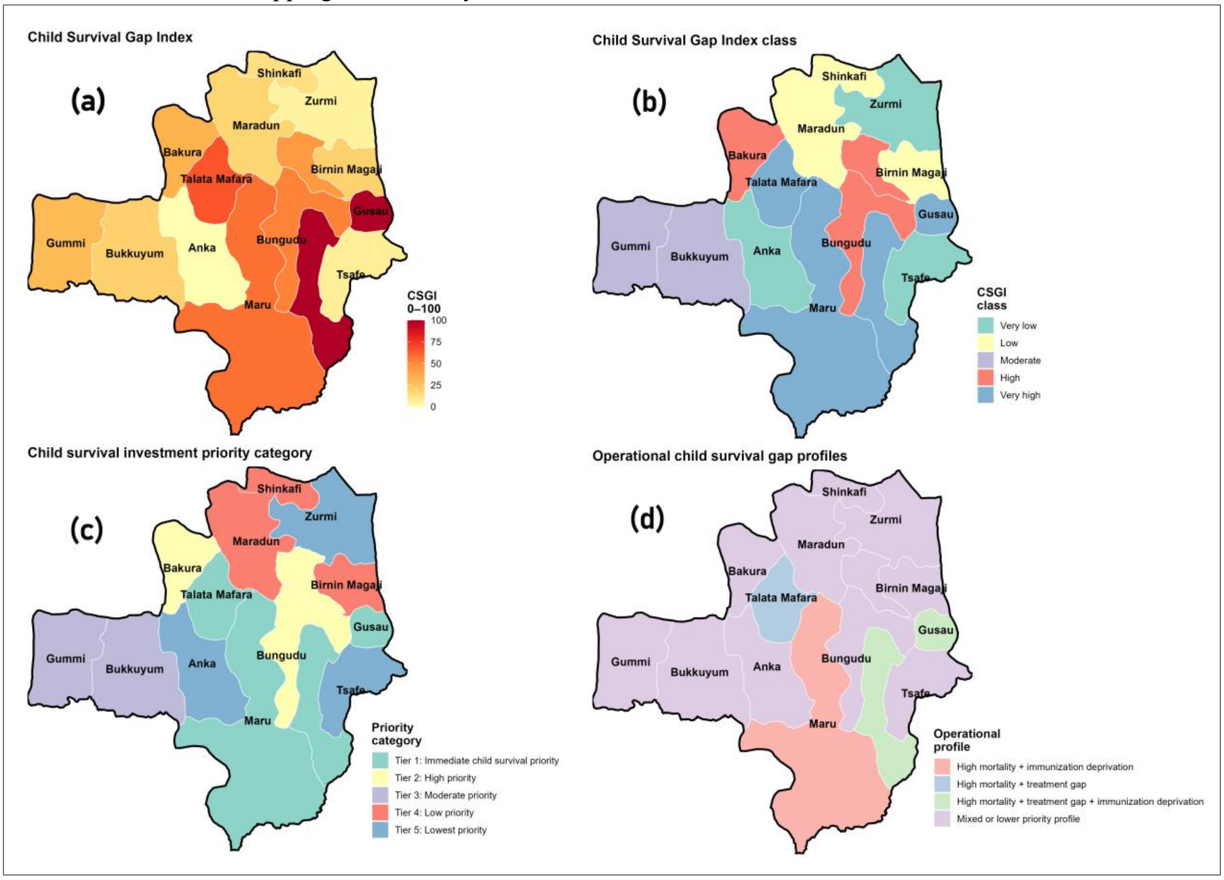

Figure 8. Spatial distribution of the child survival gap index (CSGI) and investment prioritization across Zamfara State: (a) child survival gap index (0–100); (b) CSGI classes; (c) child survival investment priority categories; and (d) operational gap profiles showing the dominant combinations of mortality burden, treatment gaps, immunization deprivation, and healthcare access constraints.

The classification of the CSGI into five categories further highlighted the unequal distribution of child survival risk across Zamfara State (Figure 8b). LGAs classified as having high or very high vulnerability formed distinct clusters in the central and eastern parts of the state, whereas LGAs with lower vulnerability were more dispersed. The investment priority map (Figure 8c) translated these vulnerability patterns into operational categories, identifying Gusau as Tier 1: Immediate child survival priority LGA. Talata Mafara, Maru, and Bungudu were also classified as high-priority investment areas requiring intensified malaria control, strengthening immunization, and improvements in healthcare accessibility.

The operational gap profiles provided additional insights into the drivers of vulnerability (Figure 8d). Several LGAs are characterized by high mortality and treatment gaps, whereas others exhibit high mortality coupled with immunization deprivation or treatment gaps combined with weak healthcare access. However, the majority of LGAs fell into a mixed or lower-priority profile, suggesting that child survival challenges in Zamfara State arise from multiple interacting determinants rather than a single dominant factor.

The correlation analysis demonstrated that the CSGI was strongly associated with its constituent components. Positive correlations were observed between the CSGI and mortality burden ($\rho = 0.775$), immunization deprivation ($\rho = 0.767$), and treatment gap components ($\rho = 0.675$). In contrast, health facility density was negatively correlated with the CSGI ($\rho = -0.741$), indicating that LGAs with fewer healthcare facilities per capita tend to exhibit greater child survival vulnerability. Nighttime light intensity showed a weaker positive association with the CSGI ($\rho = 0.385$), suggesting that urbanization alone does not necessarily translate into improved child health outcomes.

Sensitivity analysis revealed that the CSGI was relatively robust to alternative weighting schemes (Figure 9).

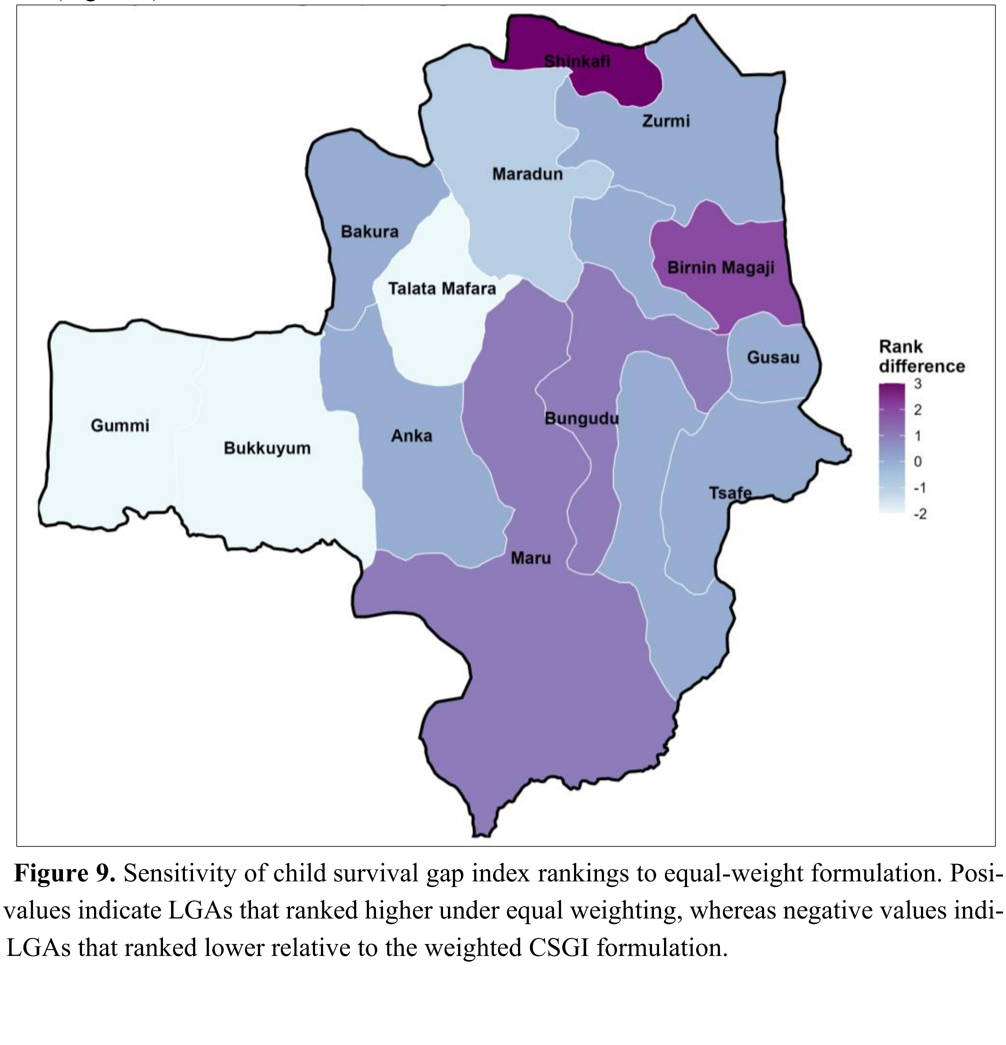

Figure 9. Sensitivity of child survival gap index rankings to equal-weight formulation. Positive values indicate LGAs that ranked higher under equal weighting, whereas negative values indicate LGAs that ranked lower relative to the weighted CSGI formulation.

Rank differences between the weighted and equal-weight formulations ranged from –2 to +3 positions across LGAs. The largest upward shift was observed in Shinkafi, which improved by three positions under equal weighting, whereas Talata Mafara, Gummi, and Bukkuyum shifted downward by two positions. Despite these minor changes, the highest-priority LGAs remained largely unchanged, indicating that the spatial prioritization of child survival investments is stable and not strongly influenced by the weighting assumptions used in constructing the index.

Overall, the CSGI provides a comprehensive framework for identifying LGAs where multiple dimensions of child survival vulnerability cooccur. By integrating malaria mortality, treatment inequities, immunization deprivation, healthcare access, and denominator uncertainty, the index offers a practical tool for evidence-based prioritization of child health interventions and resource allocation in Zamfara State.

4. Discussion

This study developed a geospatial framework to quantify underfive malaria mortality rates, evaluate denominator uncertainty, and identify child survival investment priorities across Zamfara State, Nigeria. Three major findings emerged. First, underfive malaria mortality rates exhibited substantial temporal increases and pronounced spatial heterogeneity. Second, household-enumerated and raster-derived underfive population estimates differed considerably at the LGA level despite relatively small discrepancies at the state level. Third, the Child Survival Gap Index (CSGI) revealed that multiple dimensions of vulnerability—including malaria mortality, treatment gaps, immunization deprivation, healthcare access, and denominator uncertainty—converge in a limited number of LGAs, suggesting opportunities for geographically targeted interventions.

The observed increase in underfive malaria mortality from 2014--2024 is consistent with recent evidence indicating that gains in malaria control have slowed in several high-burden African countries owing to population growth, insecticide resistance, climate variability, and persistent health-system challenges [25,26]. The concentration of predicted 2025 mortality in LGAs such as Gusau, Talata Mafara, Maru, and Bungudu further supports the hypothesis that the malaria burden is spatially clustered rather than uniformly distributed. Similar spatial concentrations of childhood malaria mortality have been reported in Burkina Faso, Uganda, and northern Nigeria, where a small number of administrative units account for a disproportionately large share of the disease burden [27,28]. These findings emphasize the need for subnational prioritization rather than uniform distribution of malaria interventions.

An important contribution of this study is the demonstration that uncertainty in underfive population denominators remains substantial even when widely adopted gridded population products are used. Although household-enumerated and raster-derived underfive populations differed by only 1.15% at the state level, discrepancies exceeded ±50% in several LGAs. This result aligns with recent studies showing that gridded population datasets perform well at broad spatial scales but may exhibit large local errors arising from settlement morphology, migration, urban growth, and modeling assumptions [29,30]. Such discrepancies have important implications for malaria programmes because estimates of intervention coverage, commodity requirements, and cost-effectiveness are highly sensitive to the denominator used [31]. In this context, operational household enumeration may provide a valuable complement to modeled population surfaces, particularly in areas undergoing rapid demographic change.

The Child Survival Gap Index provides further evidence that malaria mortality cannot be interpreted in isolation from other dimensions of vulnerability. LGAs with the highest CSGI scores were characterized not only by elevated malaria mortality but also by substantial treatment gaps, poor immunization outcomes, weak healthcare access, and high denominator uncertainty. Previous studies have shown that overlapping deprivations often amplify health risks among children, producing spatial clusters of disadvantages that are difficult to address through single-sector interventions [32,33]. The strong correlations observed between the CSGI and its mortality, treatment gap, and immunization components support this multidimensional perspective and suggest that integrated child health strategies may yield greater benefits than disease-specific approaches alone.

The operational implications of these findings are particularly relevant for malaria control programs and philanthropic organizations interested in maximizing health impact. Seasonal malaria chemoprevention (SMC), malaria vaccination, and routine immunization programs all rely on accurate target population estimates and effective geographic prioritization [34]. The identification of high-priority LGAs where multiple vulnerabilities converge offers an evidence base for optimizing resource allocation, improving intervention targeting, and strengthening programme monitoring. Moreover, the robustness of the CSGI to alternative weighting schemes indicates that the proposed framework can support decision-making even when uncertainty exists regarding the relative importance of individual components.

Several limitations should be acknowledged. First, the mortality forecasts for 2025 were derived from historical surveillance records and may not fully capture future changes in malaria transmission, intervention coverage, or environmental conditions. Second, household enumeration and raster-derived population estimates are themselves subject to measurement and modeling uncertainties. Third, the analysis was conducted at the LGA level, which may mask important within-LGA heterogeneity in malaria risk and healthcare access. Future studies should incorporate ward- or settlement-level analyses, climate and environmental drivers of malaria transmission, and Bayesian or machine-learning approaches to quantify uncertainty in both mortality predictions and population denominators [35].

Despite these limitations, this study demonstrates that integrating mortality surveillance, operational microcensus data, geospatial population modeling, and health-system indicators can provide a more comprehensive understanding of child survival vulnerability. The proposed framework offers a scalable and transferable approach for identifying priority areas for malaria and child health interventions and may contribute to more efficient and equitable allocation of resources in high-burden settings.

Data availability statement: Monthly under-five malaria mortality records (2014–2024), severe malaria cases, and severe malaria cases treated with artesunate were obtained from the Nigerian Routine Health Information System (RHIS) through the District Health Information System 2 (DHIS2) platform maintained by the Federal Ministry of Health. These data are not publicly available because they contain routine health surveillance information and are subject to national data governance and confidentiality regulations but may be made available from the relevant authorities upon reasonable request and approval.

Settlement Microcensus datasets, including household-enumerated under-five population estimates, settlement counts, household counts, routine immunization (RI) zero-dose children, not vaccinated children, and derived immunization indicators, were obtained through the Nigerian Settlement and RI Microcensus programme. These datasets are not publicly available and may be accessed through the respective implementing agencies subject to data-sharing agreements.

Raster-derived under-five population estimates for 2025 were obtained from the WorldPop project and are publicly available at https://www.worldpop.org/ and through the WorldPop Open Population Repository at https://wopr.worldpop.org/.

Health facility locations were obtained from the GRID3 Nigeria Health Facilities database and are publicly available at https://data.grid3.org/maps/GRID3::grid3-nga-health-facilities-v2-0.

Malaria transmission and burden indicators for 2024, including *Plasmodium falciparum* Parasite Rate (PfPR), Infection Rate (PfIR), Mortality Rate (PfMR), Clinical Incidence (PfIC), and Malaria Cases (PfMC), were obtained from the Malaria Atlas Project and are publicly available at https://malariaatlas.org/explorer/.

Nighttime light intensity data for 2025 were obtained from the Visible Infrared Imaging Radiometer Suite (VIIRS) Day/Night Band products provided by the National Aeronautics and Space Administration (NASA)

and the National Oceanic and Atmospheric Administration (NOAA), available at https://eog-data.mines.edu/products/vnl/.

Administrative boundary data for Nigerian states and Local Government Areas (LGAs) were obtained from the United Nations Office for the Coordination of Humanitarian Affairs (OCHA) Common Operational Datasets and are publicly available at https://data.humdata.org/dataset/cod-ab-nga.

Derived indicators generated during this study, including the Treatment Gap, Accessibility-Adjusted Treatment Gap Index (AATGI), Health Facilities per 100,000 Population, and model outputs for predicted under-five malaria mortality in 2025, are available from the corresponding author upon reasonable request.

Author Contributions: Author Contributions: Conceptualization, O.I.O., Y.J.D., and Z.I.; methodology, Z.I., O.I.O., and Y.J.D.; software, Z.I.; validation, O.I.O., Y.J.D., Y.J.C., and U.M.I.; formal analysis, Z.I.; investigation, Y.J.D., Y.J.C., U.M.I., C.A.U., O.I.O., and U.S.N.; data curation, Y.J.D., Y.J.C., and U.M.I.; visualization, Z.I.; writing—original draft preparation, Z.I. and O.I.O.; writing—review and editing, Y.J.D., Y.J.C., U.M.I., C.A.U., and U.S.N.; supervision, O.I.O.; project administration, O.I.O.; resources, Y.J.D., Y.J.C., U.M.I., C.A.U., and U.S.N. All authors have read and agreed to the published version of the manuscript.

Funding

This research received no external funding.

Informed Consent Statement

Not applicable.

Conflicts of interest

The authors declare that they have no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

Abbreviation	
CSGI	Child Survival Gap Index
AATGI	Accessibility-Adjusted Treatment Gap Index
RI	Routine Immunization
U5	Under-Five Children
LGA	Local Government Area
DUS	Denominator Uncertainty Score
PfPR	Plasmodium falciparum Parasite Rate
PfIR	Plasmodium falciparum Infection Rate
RHIS	Routine Health Information System
DHIS2	District Health Information System 2

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